

Sabbir Ahmed

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Research Experience

Research Assistant, BCS2 Lab, University of Arkansas (Remote)

[Jul 2024 - Present]

- Fine-tuned BERTweet on 8.9M tweets for DEI content classification (F1: 94.3%)
- Benchmarked the model against frontier LLMs, including Gemini 2.0/2.5 model family (Flash/Flash-Lite variants)
- Reduced API inference costs by 66% while maintaining required performance targets through selective model routing

Research Assistant, Visual Machine Intelligence Lab, JUST

[Nov 2023 - Present]

- Implemented segmentation architectures (U-Net, BASNet, SegFormer, OneFormer) for medical image segmentation
- Developed a zero-shot pipeline utilizing Foundation Models for guided unsupervised cross-domain segmentation
- Achieved a strong in-domain best-case performance of Dice 0.87 and IoU 0.77 on the validation data
- Investigating Domain Adaptation to improve cross-domain transfer in zero-shot segmentation (baseline Dice 0.28)
- Manuscript in preparation, targeting MIDL/MICCAI

Multi-Institutional Aquaculture Research Project

[2023 - 2024]

Collaborators: CSE, JUST, FMRT, Khulna University, Ministry of Fisheries and Livestock (GitHub)

- Engineered a 36-sensor ESP32-based array to monitor six critical water parameters for saltwater shrimp research
- Integrated a real-time dashboard and NVR video for remote health diagnostics
- Managed the hardware lifecycle from budget and PCB design through final deployment at the KU research facility

Publications & Academic Output

Conference Presentations:

S. Ahmed, et al. "Trustworthy AI in Agriculture: Explainable Plant Disease Symptom Pinpointing using Vision Transformers." *National Conference on Computational Intelligence and Applications (NCCIA)*, JUST, Bangladesh, 2025

S. Ahmed, et al. "Mapping and Explaining Urban Environmental Disamenities using Satellite Imagery and Explainable Vision Transformers." *IEEE Computer Society Bangladesh Chapter Symposium*, HSTU, Bangladesh, 2025

Peer Review: Served as Reviewer for IEEE COMPAS 2025 (AI, ML, and Soft Computing track), recognized for outstanding contribution.

Education

Jashore University of Science and Technology, Bangladesh

[Jun 2025 - Present]

Master of Science in Computer Science and Engineering

Jashore University of Science and Technology, Bangladesh

[2020 - 2025]

Bachelor of Science in Computer Science and Engineering

Awards and Experience

Erasmus+, MELBU Summer School, Leipzig, Germany

[Aug 2022]

- Completed 14-day intensive training on innovation management, entrepreneurship and business modeling
- Collaborated with participants from 6 countries to conduct market research with German SMEs

National Innovation Challenge, Bangladesh

[Nov 2022]

Organizers: a2i (ICT Division), Cabinet Division, UNDP Bangladesh

- Top 40 finalist out of 1,000+ nationwide applicants
- Presented to ministerial panel; received mentorship from UNDP innovation advisors

Professional Experience

Teaching Assistant, Department of CSE, JUST

[Oct 2024 - Present]

- Supporting 220+ students across 5 courses (Machine Learning, Data Analysis, DSA, Image Processing, IoT)
- Independently conducted 20+ hands-on lab sessions for ML and DSA courses with real-time grading and feedback
- Developed 35 auto-graded assessments (quizzes, midterms, finals), reducing grading time by 60%

Network Operations Intern, BdREN

[Feb - Apr 2025]

- Monitored network performance across 195 institutions (Wireshark/SolarWinds)
- Simulated routing/topologies (Packet Tracer); supported data center ops/ticketing

Technical Projects

Bengali License Plate Recognition System

[Jan - May 2024]

Technology: YOLOv8, PyTorch, Django, OpenCV

- Achieved 96.2% accuracy
- Curated dataset of annotated plates across different formats (YOLO, COCO)
- Deployed on Raspberry Pi 4; used LiDAR-triggered capture to reduce active compute time

Domain-Specific LLM for Sales Intelligence

[May 2024]

Technology: Llama-3-8B, QLoRA (4-bit), Streamlit

- Fine-tuned on 10K sales conversations using 4-bit quantization
- Efficiently answered customer queries on a consumer laptop running locally

NASA Space Apps Challenge 2023

[Aug 2023]

Technology: HTML, CSS, three.js, Python

- Led *Team_Entergalactic* to a national Top 100 ranking by developing an interactive 3D Mars visualization application
- Integrated 9 NASA APIs (Mars Rover/Helicopter, Mars Weather Data APIs)

Technical Skills

ML/Vision: PyTorch, OpenCV, YOLO

NLP: Hugging Face Transformers, Fine-tuning, LoRA/QLoRA, RAG pipelines, Prompt Engineering

Programming: Python, C/C++, Java, NumPy, Pandas, Seaborn, Git, Docker

Languages

English: IELTS 7.0

Bengali: Native

Research Interests

Zero-shot segmentation in medical imaging, parameter-efficient adaptation of foundation models (LoRA/prompt tuning), cross-modal prompting for vision-language alignment